DreamCampus — Physics Practice Workbook

Concepts, numericals, higher-level problems and a mock test

A. Concept Builder

- Explain the difference between distance and displacement with one example.
- Why is acceleration zero for uniform velocity?
- Draw a free-body diagram for a block resting on a horizontal surface.
- State the work-energy theorem in words and mathematical form.
- Explain why centripetal force is not a new fundamental force but the name for the required inward net force.
- Distinguish mass and weight.
- Why does pressure increase when the same force acts on a smaller area?
- Explain the physical meaning of specific heat capacity.
- What condition makes a motion simple harmonic?
- Differentiate electric field and electric potential.
- Why does a magnetic field do no work on a charged particle in ideal magnetic-only motion?
- Explain Lenz's law using conservation of energy.
- What is total internal reflection and what conditions are required?
- Why does increasing photon frequency increase maximum photoelectron kinetic energy above threshold?
- Differentiate intrinsic and extrinsic semiconductors.

B. Numerical Practice

- A car starts at 6 m/s and accelerates at 2.5 m/s^2 for 8 s. Find final velocity and displacement.
- A 10 kg block is pulled by 35 N while friction is 5 N opposite motion. Find acceleration.
- A 1.5 kg object moves at 4 m/s. Find its kinetic energy.
- A 500 W motor performs useful work for 40 s. Find the work done.
- A satellite moves in a circular orbit of radius r. Write its orbital speed in terms of G, M and r.
- A fluid flows through a pipe whose area changes from 4 cm^2 to 2 cm^2. If speed in the first section is 3 m/s, find speed in the second section using continuity.
- An ideal gas receives 500 J heat and does 180 J work. Find change in internal energy using the stated first-law convention.
- A spring with k = 200 N/m carries a 0.5 kg mass. Find its approximate angular frequency.
- A 10 μF capacitor is charged to 12 V. Find stored energy.
- A 4 Ω resistor is connected across 20 V. Find current and power.
- A charge q moves at speed v perpendicular to B. Write the magnetic force and describe its direction.
- A lens has focal length 0.20 m. Find its power in dioptres.

C. Higher-Level Practice

- A projectile is fired at two complementary angles with the same speed. Compare their ranges and maximum heights.
- Two blocks connected by a string are pulled on a smooth surface. Draw free-body diagrams and derive the acceleration.
- Use conservation of mechanical energy to obtain the speed of a body falling through a height h from rest.

- Explain how orbital speed and escape speed differ physically.
- Derive the continuity equation for steady incompressible flow.
- Compare isothermal and adiabatic processes in terms of heat transfer and temperature change.
- Explain resonance using a forced oscillator and identify one useful and one harmful example.
- Derive the equivalent capacitance of two capacitors in series.
- Use Kirchhoff's laws to outline a method for solving a two-loop circuit.
- Explain the path of a charged particle entering a uniform magnetic field at an arbitrary angle.
- Explain electromagnetic induction in a rotating-coil generator.
- Compare image formation by a convex lens for object positions beyond 2F, at 2F, between F and 2F, and inside F.
- Explain the photoelectric threshold frequency and stopping potential.
- Explain why binding energy per nucleon is useful for understanding nuclear stability.
- Compare diode, LED and photodiode by biasing/operation and application.

D. Mock Test — 20 Questions

- 1) A body has $u=10$ m/s and $a=2$ m/s² for 5 s. Find v.
- 2) A 3 kg body experiences 15 N net force. Find a.
- 3) Work done by a force perpendicular to displacement is what?
- 4) State the relation between momentum and impulse.
- 5) Write the expression for gravitational force.
- 6) What is the condition for SHM acceleration?
- 7) Write the wave-speed relation.
- 8) Define electric potential.
- 9) Write capacitor energy in terms of C and V.
- 10) State Ohm's law.
- 11) Write electrical power in terms of V and R.
- 12) What is the magnetic force when v is parallel to B?
- 13) State Faraday's law conceptually.
- 14) State Snell's law.
- 15) What is total internal reflection?
- 16) Write photon energy.
- 17) Write de Broglie wavelength.
- 18) What does half-life mean?
- 19) What is forward bias in a p-n junction?
- 20) Name the electromagnetic-wave region used for visible vision.

Self-check answer hints for Mock Test

1) Use $v=u+at$. 2) Use $F=ma$. 3) Zero. 4) $J=\Delta p$. 5) Inverse-square law. 6) $a=-\omega^2 x$. 7) $v=f\lambda$. 8) Work per unit charge. 9) $U=\frac{1}{2}CV^2$. 10) $V=IR$ for ohmic behavior. 11) $P=V^2/R$. 12) Zero. 13) Induced emf follows rate of change of magnetic flux and Lenz direction. 14) n■sin i=n■sin r. 15) Refraction conditions plus incidence beyond critical angle. 16) $E=hf$. 17) $\lambda=h/p$. 18) Time for half the undecayed nuclei to remain. 19) p-n junction biased to reduce depletion barrier. 20) Visible.

Recommended practice cycle

Day 1: Concept Builder → Day 2: Numerical Practice → Day 3: Higher-Level Practice → Day 4: Mock Test → Day 5: error-log revision. Repeat weak chapters.